



JERUSALEM COLLEGE OF ENGINEERING, CHENNAI – 600 100
(An Autonomous Institution Affiliated to Anna University, Chennai)

Reg. No. :

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Question Paper Code : 135156

B.E. DEGREE END SEMESTER EXAMINATIONS, NOV/DEC 2025 & JAN 2026

Fifth Semester

Computer Science and Engineering (Artificial Intelligence and Machine Learning)

JAL2521 - TIME SERIES ANALYSIS AND FORECASTING

(Regulation 2023)

Time : Three hours

Maximum : 100 marks

Answer ALL questions

PART A – (10 × 2 = 20 marks)

1. What is an autoregressive model?
2. What is meant by forecast horizon?
3. What are forecast and residual errors?
4. What are the ways to monitor forecasting model performance?
5. Differentiate prediction interval and confidence interval.
6. List the different methods used for variable selection in regression.
7. What are first order and second order moving average models?
8. Can a non-stationary process be converted into a stationary process? If yes, mention how it can be done.
9. What is forecast pooling?
10. What is backpropagation in neural networks, and what is its primary purpose?

PART B – (5 × 16 = 80 marks)

11. (a) Discuss the internal structure of a time series. Explain trend, seasonality, cyclical, and irregular components with suitable illustrations

Or

11. (b) Compare and contrast qualitative and quantitative forecasting methods with examples.
12. (a) Discuss the numerical characteristics of time-series data. Explain how measures such as mean, variance, autocovariance, and autocorrelation describe the internal structure of a time series.

Or

12. (b) (i) Discuss how the performance of a forecasting model is evaluated. (8)
- (ii) Explain the role of accuracy measures such as Mean Absolute Error, Mean Squared Error, and Mean Absolute Percentage Error. (8)
13. (a) Discuss the F-test for significance of regression. Derive the test statistic and explain how it is used to evaluate the overall adequacy of a regression model.

Or

13. (b) Discuss generalized and weighted least squares methods.
14. (a) Explain how Autocorrelation Function (ACF) and Partial Autocorrelation Function (PACF) are used to identify AR and MA models. Illustrate the distinct ACF and PACF patterns for AR(p), MA(q), and ARMA(p, q) models.

Or

14. (b) Discuss in detail the ARIMA(p, d, q) model. Explain how autoregressive, differencing, and moving-average components are combined in ARIMA modeling with an example of each.
15. (a) Explain in detail about the multivariate time series models.

Or

15. (b) Explain the Bayesian approach to forecasting. Describe how prior information is combined with observed data using Bayes' theorem to update forecasts.
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